

STATEMENT PROFILE

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REVIEW

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1. PURPOSE & SCOPE

This statement sets out the position of Advanced Invasives Ltd on how Invasive Knotweeds (Japanese Knotweed *Reynoutria japonica* and related taxa) came to be present within UK infrastructure corridors, and on the mechanisms by which they continue to spread along them.

It is issued as public information for infrastructure owners and managers, their advisers and contractors, and for anyone assessing how a Knotweed stand in a rail, road or canal corridor is likely to have arisen. The statement addresses historical introduction and contemporary spread. It does not set out control methodology, which is covered separately, and it does not constitute legal advice.

THE PRACTICAL CONSEQUENCE

Where an Invasive Knotweed stand is found along a linear corridor, the available evidence does not support an assumption that it was deliberately planted there. The more probable explanation is introduction from an adjacent source, followed by redistribution along the corridor through disturbance.

2. SUMMARY POSITION

- **Spread is clonal, and disturbance drives it:** In the UK, Invasive Knotweeds spread almost entirely by vegetative dispersal rather than by seed, and both anthropogenic and natural disturbance encourage that spread (Jones et al. 2018).
- **The invasive clone arrived after the network was built:** the Japanese Knotweed clone that invaded Britain became commercially available in Europe only in 1848 and reached Kew in August 1850 (Bailey and Conolly 2000), which is after the principal period of British main-line construction in the 1830s and 1840s. It cannot have been used to stabilise embankments that were already built.
- **There is no evidence of planting as a matter of policy:** We are not aware of any documentary evidence that Invasive Knotweeds were planted along railway embankments as a matter of company policy, and the contemporary horticultural record of railway bank planting does not name them. Engineers of the construction period did specify what was planted, Isambard Kingdom Brunel among them on the Great Western Railway, which gives that absence some weight, although section 4.6 sets out why we do not rest our position on it.
- **Ornamental planting by individuals is documented, but it was individual:** Railway banks were planted ornamentally from the 1880s onwards, but the record shows this was driven by station staff, adjacent landowners and voluntary associations rather than by the railway companies, and contemporaries described such efforts as unusual.
- **Garden escapes and cast-outs are an important historical source:** These have contributed Invasive Knotweed to the rail and road network from the Victorian era to the present day, particularly near urban and historically industrialised areas (Conolly 1977).
- **Contemporary spread is a maintenance phenomenon:** Linear infrastructure corridors are high disturbance environments, and following establishment, further spread along them arises principally from necessary maintenance activity rather than from historical planting.

3. HOW INVASIVE KNOTWEEDS SPREAD

3.1 Vegetative dispersal and disturbance

In the UK, Invasive Knotweeds are spread primarily through vegetative, that is asexual or clonal, dispersal, and not by seed. Spread is encouraged by anthropogenic disturbance and by natural disturbance processes such as flooding (Jones et al. 2018). Once Invasive Knotweed is present in an area, disturbance by people, some animals, natural processes, and by certain physical control treatments used in its management all have the potential to produce regenerative material.

3.2 Mechanisms along linear infrastructure

Disturbance and mismanagement are of particular importance in riparian habitats and along linear infrastructure such as canals, roads and railways, because of the potential in those settings for long-distance dispersal (Cottet et al. 2020). We identify three distinct mechanisms, which are frequently conflated, but which differ in what they move and how far.

- **Vegetation management moves aboveground material:** Cutting or mowing aboveground parts of the plant can result in spread, particularly where flail mowing is used, which produces many small fragments of cut stem (Colleran and Goodall 2015; Cottet et al. 2020; Jones et al. 2020). In 2021 the Esk and Tay Invasive Plant Project, part of the Scottish Invasive Species Initiative, recorded cut Bohemian Knotweed stems that had subsequently re-rooted.
- **Groundworks move belowground material:** The movement of soil containing rhizome fragments during redevelopment, highway construction and landscaping was historically the primary driver of spread between and beyond river catchments in the UK (Palmer 1990). Roadside and trackside engineering works, including cabling, drainage and carriageway widening, disturb and transport rhizome. This mechanism is distinct from verge cutting, and arguably the more consequential of the two, because rhizome is the more reliable regenerative unit and excavated material is frequently moved considerable distances along the corridor. The same applies to canal, river and rail corridors.
- **Air movement redistributes broken material along the trackside:** On operational railways, the speed of passing trains creates the potential for broken stem material to be carried along the trackside in slipstreams; this has also been described alongside roads for other invasive species.

4. THE HISTORICAL RECORD

4.1 Boundary planting and the Railway Regulation Act 1842

The importance of establishing clear boundaries between the railway and adjacent land was recognised early. The Railway Regulation Act 1842 required public access to the railway to be restricted on safety grounds, and therefore required the railway companies to maintain fences along their lines; in more rural areas thorny hedges, often Hawthorn (*Crataegus monogyna*), were planted and left to form an impassable barrier, creating the largely empty boundary corridors that still flank the network. Scotch Broom and Common Gorse were sown on embankments, presumably to stabilise them (Garden History Girl 2022).

Planting on the new lines was a specified matter rather than an afterthought. Isambard Kingdom Brunel specified plantings within his designs for the Great Western Railway, and the species named for railway land in this period are the boundary hedge, principally Hawthorn, with Scotch Broom and Common Gorse on the embankments. We have found no Invasive Knotweed named in that context. That absence is evidence of a kind, though it is not conclusive, for the reasons set out at section 4.6.

Newly built sections of railway were quickly colonised by Couch Grass, which bound the freshly turned soil but created a fire hazard from engine sparks and cinders, requiring periodic clearing or controlled burning along the track (Garden History Girl 2022). The corridor was therefore a high disturbance environment from the outset.

4.2 Chronology: the invasive clone arrived after the embankments

The literature has long carried two conflicting introduction dates, 1825 and the late 1840s, and Bailey and Conolly (2000) resolve them. Both are in a sense correct, but only one is relevant.

A plant answering to Japanese Knotweed was indeed grown at the Horticultural Society's Chiswick garden from about 1825, where it had been mistaken for *Houttuynia cordata* and planted in an artificial swamp; it failed to thrive and flowered only once. That accession came from China, and Chinese plants are morphologically distinguishable from the Siebold clone. Bailey and Conolly conclude that even if the

Chiswick plant was ever distributed to any extent it is doubtful that any has survived, and that it “has had no discernible role in the Japanese Knotweed invasion of Europe”.

The plant that did invade is von Siebold’s Japanese Knotweed clone, propagated and heavily promoted from his nursery at Leiden. It was awarded a gold medal in 1847, and the first date on which it was commercially available in Europe is, in Bailey and Conolly’s words, undeniably 1848. The Royal Botanic Gardens Kew “Inwards Book” records *Polygonum sieboldii* received on 9 August 1850 as part of a large batch from Siebold of Leiden.

The principal period of British main-line railway construction, and therefore of embankment building, was the 1830s and 1840s. The invasive clone was not available in Britain at this time. A plant that reached Kew in 1850 cannot have been specified to stabilise embankments that were already built, and the 1825 date, which is the one usually produced in support of the railway story, refers to a different accession that played no part in the invasion.

Von Siebold’s catalogue promoted the plant as able to “provide shelter for young plantations and consolidate sandy hills and dunes”, and a claim of satisfactory trials stabilising slopes along railroad tracks and sandbanks is also attributed to the catalogue (Del Tredici 2017). However, there is no evidence that the practice was taken up in Britain (Bailey and Conolly 2000).

4.3 The earliest records along roads and railways

The earliest railway-associated record we are aware of comes from Aberdeen, where Trail (1899) had been observing a piece of waste ground formed by the filling in of an old channel of the River Dee in 1896. This land was subsequently used for railway sidings, and three stems of Japanese Knotweed were noted there in 1898 (cited in Conolly 1977).

Between 1902 and 1904 there are several records from the vicinity of Edinburgh, all from sites on waste ground, by railways or in nursery gardens. In 1903 Melville observed its spread near Church Stretton, Shropshire, into the roadside habitat, mingling with the native vegetation; we note that Conolly records a poor specimen of Melville’s from Church Stretton as more probably Giant Knotweed, so that particular record should be treated with caution (all cited in Conolly 1977).

Each of these postdates the construction of the lines and roads concerned by several decades.

4.4 Industrial planting: a plausible route, and not the railway companies

There is one route by which deliberate planting may well have occurred, and it is worth setting out because it is more plausible than the railway story and is frequently confused with it.

The first certain naturalisation of Japanese Knotweed in the British Isles is on cinder tips at Maesteg (Storrie 1886, cited in Conolly 1977), which is not the profile of an ordinary garden escape. Maesteg in the early to middle nineteenth century sat at the heart of the Industrial Revolution and was home to the Llynfi and Maesteg iron works, with their blast furnaces, coke ovens and extensive tips. Von Siebold’s catalogue promoted the plant for the consolidation of sandy hills and dunes, and the extension from sand dunes to slag heaps is not a large one. The association with industrial ground is not confined to Britain: early German records point to the use of the plant as a soil stabiliser in Westphalia, where it was also used as fodder (Krausch, fide Starfinger), and Conolly records Schemmann (1884) noting the species as long established as an escape near a coal mine on the banks of the Ruhr.

The ecology supports the inference. Conolly observes that the readiness to colonise cinder tips, railway ballast and similar well-drained pioneer habitats is not surprising for a species which in its native Japan is characteristically an early colonist of volcanic soils and occurs even on the sulphurous soils of fumaroles with a pH below 4. A plant that evolved as a first coloniser of volcanic ash fields had little to fear from anything a steel works or a colliery could produce, and it is still put to precisely this use in Japan, although there the Knotweed canopy is accompanied by co-occurring species that also bind the soil, which is not the case in its invasive range.

We should be clear that this is a hypothesis: we are not aware of any document showing an ironmaster specifying Invasive Knotweed for a tip. It is proposed because it is the most plausible deliberate-planting route in the British record; it accounts for the earliest naturalisation better than a garden escape, and because industrial sites and their tips frequently sat adjacent to the rail network, which is one way in which an apparently railway-planted stand may have arrived without any railway company having planted anything.

4.5 The ornamental planting movement, 1880s to 1920s

Railway banks and stations were planted ornamentally during this period, and the movement is well documented in the horticultural press. In 1884 the Midland Railway began a best-kept station garden competition for its staff, and other companies followed. Railway companies rented land adjacent to stations to employees for gardens and allotments (Garden History Girl 2022).

In 1900 Lord Grey and Ellen Willmott formed the Railway Banks Floral Association to promote the planting of station gardens and railway embankments, with its first branch in Northumberland. Willmott offered to send “as many plants, seeds and bulbs as can be distributed and planted”, and in November 1906 wrote in *The Garden* with practical recommendations for sowing flower seeds in bare spots, planting Crocus and Daffodil bulbs in grass, and planting herbaceous perennials and shrubs. *The Cyclopaedia of American Horticulture* (1902) described the Association as aiming to engage the interest of “owners of adjacent properties” and to collect “money and materials for sowing and planting railway banks and cuttings” (all cited in Garden History Girl 2022).

In 1903 Theodore Walker, a Leicestershire landowner, wrote to *The Garden* advising that local gentry were working with the local stationmaster and porter to plant up the railway banks and to provide them with plant material, seeing opportunities for “beautifying our railway banks and adding to the interest of railway travel”. In 1924 Gertrude Jekyll wrote in *The Garden* that station masters were “public benefactors” whose planting served to “refresh the weary traveller” (cited in Garden History Girl 2022).

Two features of this record matter for the present question. First, the activity was driven by station staff, adjacent landowners and voluntary associations, and not by the railway companies as a matter of policy; the companies’ own contribution was competitions, prizes and the letting of land. Second, the planting was concentrated at and around stations rather than distributed along the open line.

4.6 What the horticultural record does and does not name

The species recorded in this literature are named repeatedly and specifically. Contemporary accounts of railway bank and station planting name lilies, irises, gladioli, delphiniums, phloxes, carnations, fuchsias, Michaelmas daisies, *Anemone japonica*, rhododendrons, conifers, narcissus, tulips, peonies, roses, crocus and daffodils (Garden History Girl 2022).

Across this body of horticultural writing from the 1870s to the 1920s, we have found no reference to the planting of Invasive Knotweeds on railway banks. However, the gardening authorities of exactly this period were recommending Knotweed for rough ground. William Robinson listed Japanese Knotweed in *The Wild Garden* (1870) under plants with large or graceful foliage suitable for naturalisation, and in *The English Flower Garden* (1898) wrote that the great Japan Knotworts are handsome in rough places in the wild garden and that there is no better plant for semi-wild places; Gertrude Jekyll, whom we quote above on station gardens, elsewhere envisaged a woodland walk flanked by the great Knotweeds of Japan (all in Bailey and Conolly 2000). The people planting railway banks were following those authorities. The absence of Knotweed from the railway-bank literature is therefore weaker evidence than it first appears.

Bailey and Conolly document a substantial nationwide trade in Knotweed plants, with nurseries well distributed throughout the British Isles and an efficient postal service to serve them; Siebold quoted a price per hundred plants, Gauntlett’s and others sold Giant Knotweed by the dozen, and Japanese Knotweed remained on sale for at least 87 years. A plant sold in those quantities for that long does not need to have been planted by anyone in particular to end up everywhere.

The comparator case is instructive. Oxford Ragwort (*Senecio squalidus*) is the plant most strongly associated with the expansion of the railway network, having escaped from the Oxford Botanic Garden and spread along the lines (Garden History Girl 2022). It did so by wind-dispersed seed and without being planted by anyone. An invasive plant may therefore be strongly associated with the railway network without the railway companies having introduced it.

4.7 Where the plant actually sits on the network

Conolly’s observations of distribution provide independent corroboration. Around metropolitan London she records an abundance of stands along the railway embankments within the densely built-up area, but notes that further out these become restricted to the immediate vicinity of stations or goods yards: on the old L.M.S. line out of St Pancras, inter-station stands are common as far as Elstree, but further north they are conspicuous only at St Albans, Luton, Bedford and then Leicester, with the same pattern on the Southern lines out of Victoria and Waterloo changing at about East Croydon and Weybridge respectively (Conolly 1977).

Planting as a matter of policy would produce stands distributed along the line. Clustering at stations and goods yards is what garden escapes, tipping and goods traffic produce, and it matches the horticultural record set out above, in which ornamental planting was concentrated at and around stations rather than along the open line.

4.8 A documented example of individual sowing, and what followed

One documented case illustrates both that ornamental sowing along the open line did occur and how unusual it was considered. The Rev. Prebendary John Frederick Chanter was rector of Parracombe, on the Lynton and Barnstaple Railway, from 1886 until 1916. His obituary in the *Transactions of the Devonshire Association* records that:

“Soon after the Barnstaple and Lynton Railway was opened, he conceived the notion of scattering flower seeds along the sides of the track, a practice which, however reprehensible from the botanist’s point of view (though the botanist can hardly have been deceived) certainly added a novelty to the journey, and an embellishment not often imitated.”

Devonshire Association (1939)

The line opened in 1898, so the sowing took place around the turn of the century. The passage confirms that individuals sowed ornamental material along the open line, that such sowing was the act of a private individual rather than the railway company, and, in the phrase “not often imitated”, that a contemporary regarded the practice as uncommon. Chanter scattered seed, and Invasive Knotweed in Britain does not spread by seed in any significant way.

Interestingly, however, in Conolly’s discussion of Himalayan Knotweed (*Polygonum polystachyum*, now *Koenigia polystachya*) she writes that the finest stands of naturalised plants in Britain must be along the old railway from Barnstaple to Lynton, around Woody Bay and by the River Heddon at Parracombe. That is the line Chanter sowed and the parish he served. The earliest authentic evidence of escape for that species is 1917, from a correspondent at Lynton, with Druce (1929) recording it in great abundance near Woody Bay and along the railway from Barnstaple to Lynton.

We do not consider this to show that Chanter introduced it, and two things weigh against that reading: Conolly records that most occurrences of Himalayan Knotweed derive from garden plantings that have persisted in neglected ground, or from material dumped from nearby gardens, rather than from sowing; and Lynton itself had gardens, the first report coming from a correspondent there. What the coincidence does show is that the open line was a receptive habitat, that ornamental activity and subsequent Knotweed establishment can occur in the same place without one demonstrably causing the other, and that correlations of this kind are precisely what make the origin question difficult to settle from the historical record alone.

4.9 Garden escapes and cast-outs

Garden escapes and cast-outs have formed an important source of Invasive Knotweed throughout the rail and road network from the Victorian era through to the present day, and particularly near urban and historically industrialised areas (Conolly 1977). Bailey and Conolly (2000) find that a high proportion of isolated stands in England and Wales are associated with local estates and gardens, but note that there are many others “in quite other habitats, not associated with estates but naturalised on roadsides, railway tracksides or banks, for which some other explanation is called for”. This statement is an attempt at that explanation. It is likely that some private landowners also planted Invasive Knotweeds to stabilise slopes and embankments near railways, with those plants subsequently spreading along the network; the industrial case set out at section 4.4 is the strongest form of that argument. Neither has been officially documented, and both should be treated as plausible mechanisms rather than established ones.

4.10 Maintenance as the dominant contemporary mechanism

Railways remain high disturbance environments. Following establishment, Invasive Knotweed spread along them is therefore likely to be due to necessary maintenance processes such as ballast recycling and trackside vegetation management, together with the groundworks and air movement mechanisms described in section 3.2, rather than to any historical decision to plant.

5. WHAT THIS MEANS IN PRACTICE

IMPLICATION	WHAT IT MEANS FOR INFRASTRUCTURE MANAGERS
Vegetation management method matters	Cutting and flail mowing generate regenerative material, so a maintenance regime intended to keep a corridor clear may be extending the stand it is cutting (Jones et al. 2020).
Groundworks need separate consideration	Cabling, drainage and widening move rhizome, and a biosecurity regime that does not consider this will fail.
Arisings and ballast are pathways	Material moved along a corridor during maintenance moves Invasive Knotweed along it too, which is why spread on linear infrastructure tends to follow the corridor rather than radiate from a point.
Riparian and linear settings carry long-distance risk	Both concentrate disturbance and the transport mechanism in the same place (Cottet et al. 2020).
Historical planting should not be assumed	Where the origin of a stand is contested, neither the chronology nor the documentary record supports a default assumption that infrastructure owners planted it.

6. LIMITATIONS OF THIS STATEMENT

The historical record for ornamental planting along the UK rail network is incomplete, and much of it survives in horticultural periodicals and secondary accounts rather than in company records. Garden History Girl (2022) is a secondary, non-peer-reviewed source; it is used here for garden history context and for its rendering of primary material in *The Garden*, *Revue Horticole* and *The Cyclopedia of American Horticulture*, and not for any claim about Invasive Knotweed specifically. Where practicable those primary sources should be consulted directly before the statement is relied upon in a contested setting.

The absence of Knotweed from the railway-bank literature is qualified at section 4.6 by the fact that the leading gardening authorities of the period recommended it for rough and semi-wild ground, and our position does not rest on that absence alone. Absence of documentation is not proof that planting did not occur. Our position is that the chronology of construction rules out Invasive Knotweed as a construction-period stabilisation species for most of the network, and that the available evidence does not support deliberate planting as a matter of policy thereafter. Both are narrower claims than asserting that no Invasive Knotweed was ever planted alongside a railway.

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Records attributed above to Schemmann (1884), Storrie (1886), Hooker (1880), Trail (1899), Melville (1904), Druce (1929) and Synge (1956) are cited from Conolly (1977), which remains the authoritative account of the early British record and which we have consulted directly.

ACKNOWLEDGEMENT

This statement rests heavily on Ann Conolly's 1977 paper in *Watsonia*, which remains the foundation of what is reliably known about the early British invasion history of these species. It was the product of years of meticulous work through herbaria across the country and through the relevant, and sometimes not so relevant, literature, and her four decadal distribution maps still frame the subject. Much of the argument above is simply her record, read carefully.

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